

Advanced Customer Churn Prediction

Optimizing Retention with XGBoost & Class Weighting

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The Business Problem



Retail Reality

Acquiring new customers takes 5x more budget than retention



Data Challenge

Real-world dataset: 73% Stay vs 27% Leave



Our Goal

Identify at-risk users without majority class bias

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Architecture Overview

Technology Stack

- Python
- Pandas
- Scikit-learn
- XGBoost

Pipeline Stages

Data Ingestion → Preprocessing → Advanced Modeling → Deployment

Key Innovation

Moved beyond basic algorithms to tuned XGBoost pipeline with intelligent class weighting

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Why XGBoost?

Random Forest Failure

Extreme overfitting on tabular dataset

XGBoost Advantages

- Superior tabular data performance
- Built-in L1/L2 regularization
- Prevents noise learning

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Handling Class Imbalance

Attempt 1: SMOTE

Result: 97% training accuracy

Created synthetic data → Overfitting

Attempt 2: Downsampling

Result: 74% generic accuracy

Removed valid data → Information loss

Final: Class Weights

scale_pos_weight: ~2.77

Penalize missing churners 2.77× more

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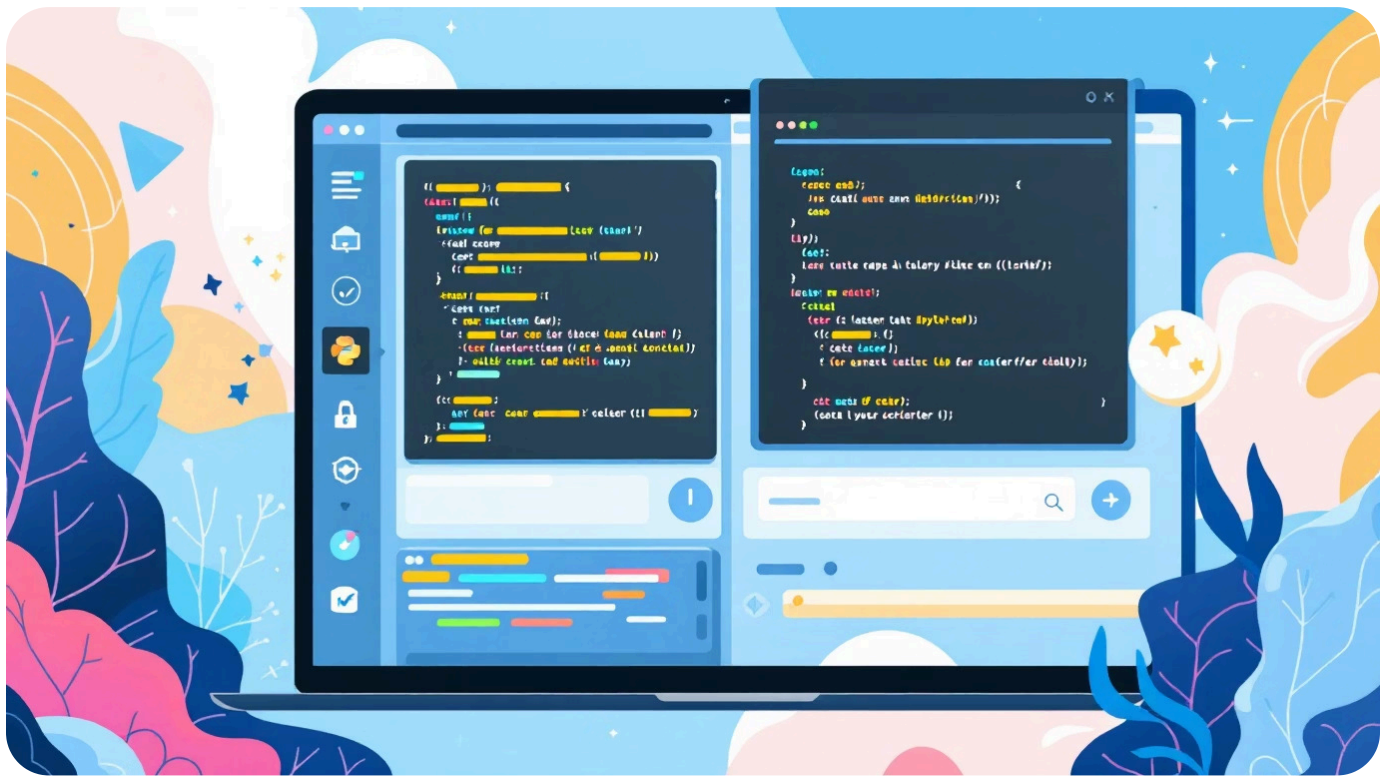
Hyperparameter Tuning

Method

GridSearchCV tested 36 unique combinations

Winning Parameters

- Learning Rate: 0.1
- Max Depth: 5
- N Estimators: 100



Result

Automated search for optimal modeling sweet spot

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Fighting Overfitting

Before

Train 97% vs Test 78% ~19% gap



After

Train 92% vs Test 85% ~7% gap

The Problem

Train: 97% vs Test: 78%

~19% generalization gap

The Fix

- Limited max tree depth to 5
- Applied subsampling (0.8)

Outcome

Gap reduced to ~7%

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Validation Strategy

Stratified K-Fold

3-fold cross validation

Enforced Balance

Every fold: 27% churners

Robust Results

Prevents lucky splits

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Performance Metrics

76%

Accuracy

Stable, robust performance

76%

Churn Recall

3 of 4 leaving customers

53%

Precision

Acceptable trade-off

Business impact: Retention team can target right users before they leave

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Deliverables & Next Steps

Production Ready

- Trained Model (.pkl)
- Encoding Rules
- Evaluation Report

Future Roadmap

1. Feature engineering: Tenure → Years
2. Real-time Web API deployment
3. Continuous monitoring dashboard

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